Simple Chatbot

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What is a chatbot?

- A chat bot is a **software encoded robot** that is able **to respond to messages** that a user sends
- Chatbots are usually found on online shops for faster customer service responds towards template questions
- A sample chatbot that has been commercially used as an application is SimSimi

Data Acquisition

- Just like topic modelling or text generation, chatbots require data
- For what you might ask? Data in chatbots provide the answers to template questions a user might ask may it be for random questions or information gathering purposes
- The data we are going to use are texts that contain information about random topics that we want to discuss. For example, a food oriented chatbot will contain data regarding its menu, discounts, promo’s etc.

Data Acquisition

- This will be our sample dataset, the default data that will be given to everyone discussing about Active Learning
- Take 10-15 minutes to gather text data discussing about machine learning and add it to this file.

Mod10.txt - Notepad

File Edit Format View Help

Active Learning (subset of machine learning) is a kind of query based learning
where the user queries for specific information based on a certain region of interest.

The fundamental assumption behind the active learner algorithm concept is that an ML
algorithm could potentially reach a higher level of accuracy while using a smaller
number of training labels if it were allowed to choose the data it wants to learn
from.

Building CB based on AL
Instead of storing each instance in the memory the learner can exploit from the
hypothesis postulated by AL that user has the liberty to choose the data to learn
from specific to the learning algorithm.
So in comparison to regular ML algorithms where we randomly picked up the
instances we can employ AL to do it more smartly in an interactive approach.

examples for AL
a)Speech recognition; Accurate labeling of speech utterances is time consuming
and require expertise, annotation at word level can take a ten times longer than
actual audio, and annotating phonemes can take 400 times as long,

b)Document classification and filtering: Learning to classify documents or any
kind of media requires that users label each document or media file with particular
labels, like relevant or irrelevant

Wordnet

- WordNet is a lexical database for the English language, which was created by Princeton, and is part of the NLTK corpus.
- You can use WordNet alongside the NLTK module to:
  - Find the base word (WordNet.lemma)
  - meanings of words
  - Synonyms, antonyms, (WordNet.Synsets)
  - and more
- Let's cover some examples.

We will now use that data and wordnet to make our own chatbot

Google Colab Code